

Solution Requirements

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

This document lists what the EduGenie solution does and how well it performs.

- Functional Requirements (FR) define specific behaviours, functions or features of the system (what the system should do).
- Non-Functional Requirements (NFR) specify the criteria used to judge how the system operates, rather than specific behaviours (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
1	Authentication	The current release is open-access: no sign-up or login is needed, so any learner can start using EduGenie immediately. Account-based login is planned as a future enhancement.	Low
2	Authorization levels	All five features (Ask, Explain, Summarize, Quiz, Learning Path) are available to every user through a single learner role. Role-based access (Student vs. Educator) is planned for the future.	Low
3	External interfaces	The system must connect to the Google Gemini API through the Google GenAI SDK for question answering, quizzes, summaries and learning paths, and run the local LaMini-Flan-T5-783M model through the Hugging Face Transformers library, falling back to Gemini if the local model cannot load.	High
4	Transaction processing	Each request must be handled as a simple HTTP call: GET for /qa and /learn/recommendations, and POST with a JSON body for /explain, /summarize and /quiz. Blocking AI calls must run in a thread pool so the server stays responsive to other users.	High
5	Reporting	The quiz must show 3 multiple-choice questions with 4 options each and, on request, an instant Correct or Incorrect message with the right answer. Answers, summaries and learning paths must be displayed as readable formatted text (headings, bold text and bullet lists).	Medium
6	Business rules, etc.	Empty input must be rejected with a clear message (HTTP 400). A quiz is accepted only if each question has exactly 4 options and the answer matches one of them. Prompts must instruct the AI to act as a tutor and answer in simple, student-friendly language.	High

7	Compliance to laws or regulations	Prompts keep the AI focused on educational content, and the Gemini API's built-in safety filters apply to every cloud request. The application collects and stores no personal data.	Medium
S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
8	Other	The web page must be served from Jinja2 templates with static CSS and a small amount of plain JavaScript, with no heavy front-end framework, so it loads quickly. Errors must appear in the page instead of crashing it.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages must open quickly, and AI features must return a result in a reasonable time. The first Explain request is slower because the local model (about 3 GB) is downloaded once.	Page loads in under 2 seconds; Gemini-based features respond in about 10 seconds or less under normal conditions
2	Scalability	The asynchronous FastAPI backend must handle many learners at the same time without dropping requests, and the Vercel deployment must scale with demand.	Several simultaneous users served without failed requests
3	Security & Data Privacy	The Gemini API key must be kept in environment variables (.env locally, Vercel settings online), listed in .gitignore and never committed to GitHub. AI text must be escaped before it is shown so it cannot inject HTML. No personal data is stored.	Zero credentials in the repository; no user data stored
4	Reliability & Availability	The application must stay up when the AI service fails or the input is invalid. Errors are returned as clear JSON messages and shown to the user, and the local explanation model falls back to Gemini automatically.	No crashes on invalid input or API errors; every failure shows a readable message
5	Usability & Accessibility	The interface must be simple (one dropdown, one text box, one button), responsive from phones to desktops, high-contrast, with labelled controls and visible keyboard focus.	Usable at 360 px width; all controls reachable by keyboard
6	Other (Maintainability & Portability)	The code must be modular, with one module per feature, and run on Windows, Linux and macOS with Python 3.10 or newer.	Same code runs on all three operating systems; one file per feature

Note: Authentication, role-based authorization and data storage are intentionally out of scope for the first release, which keeps the tool free and instant to use. They are listed as future enhancements.